TIØ4145 Corporate Finance Chapters 12 and 13

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Time for a Quiz!

- https://kahoot.it/

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Estimating the Cost of Capital

- Chapter 12
  - 12.1 The Equity Cost of Capital
  - 12.2 The Market Portfolio
  - 12.3 Beta Estimation
  - 12.4 The Debt Cost of Capital
  - 12.5 A Project's Cost of Capital
  - 12.6 Project Risk Characteristics and Financing
  - 12.7 Final Thoughts on the CAPM

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Learning objectives

- Estimate a company's **cost of capital using the CAPM** equation for the Security Market Line.
- Describe **the market portfolio** and how it is constructed in practice.
- Describe **common proxies** for the market return and the risk-free rate.
- Define **alpha and beta** and explain how they are generally **estimated**.
- Compare the **use of average return versus beta** and the SML to estimate cost of equity capital.
- Estimate the **cost of debt**,
  - given a company's yield to maturity, probability of default, and expected loss rate.
  - given a company's debt beta, the risk free rate, and the market risk premium.
- Discuss the **difference between the yield to maturity and the cost of debt** when there is low versus high default risk.
- Illustrate the use of comparable companies' **unlevered betas or unlevered cost of c**apital to estimate project cost of capital.
- Discuss **strengths and weaknesses of the CAPM**.

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The Equity Cost of Capital

- Cost of Capital
  - the best expected return available in the market on investments with similar risk.
- Capital Asset Pricing Model (CAPM) provides a practical way to identify an investment with similar risk.
- The cost of capital of any investment opportunity equals the expected return of available investments with the same beta.
- The estimate is provided by the **Security Market Line equation:**

$$E(R_i) = r_f + \underbrace{\beta_i \times \left(E(R_m) - r_f\right)}_{Risk\ Premium\ for\ Security\ i}$$

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Application of CAPM

- Investment in a firm's stock:
  - To value a share of stock, we need to calculate the equity cost of capital.
  - To implement the CAPM equation we need a number of key inputs:
    - Construct the market portfolio, and determine its expected excess return over the risk-free interest rate.
    - Estimate the stock's beta.

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The Market Portfolio

- How is the market portfolio constructed?
- What are common proxies used to represent the market portfolio?
- How can we estimate the market risk premium?

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Construction of the market portfolio

- The investment in each security is proportional to its market capitalization.
- The total market value of a firm's outstanding shares:

$$MV_i = (\text{Number of Shares of } i \text{ Outstanding}) \times (\text{Price of } i \text{ per Share}) = N_i \times P_i$$

- Value-Weighted Portfolio

$$x_i = \frac{\text{Market Value of } i}{\text{Total Market Value of All Securities}} = \frac{MV_i}{\sum_j MV_j}$$

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Market Indexes

- Common Proxies for Market Portfolio
  - Market Indexes
    - Report the value of a particular portfolio of securities.
  - Examples:
    - Europe: Euro Stoxx 50, MSCI EMU, MSCI Europe
    - U.S.: S&P 500, Wilshire 5000, Dow Jones Industrial Average (DJIA)
- Most practitioners use the S&P 500 as the market proxy, even though it is not actually the market portfolio

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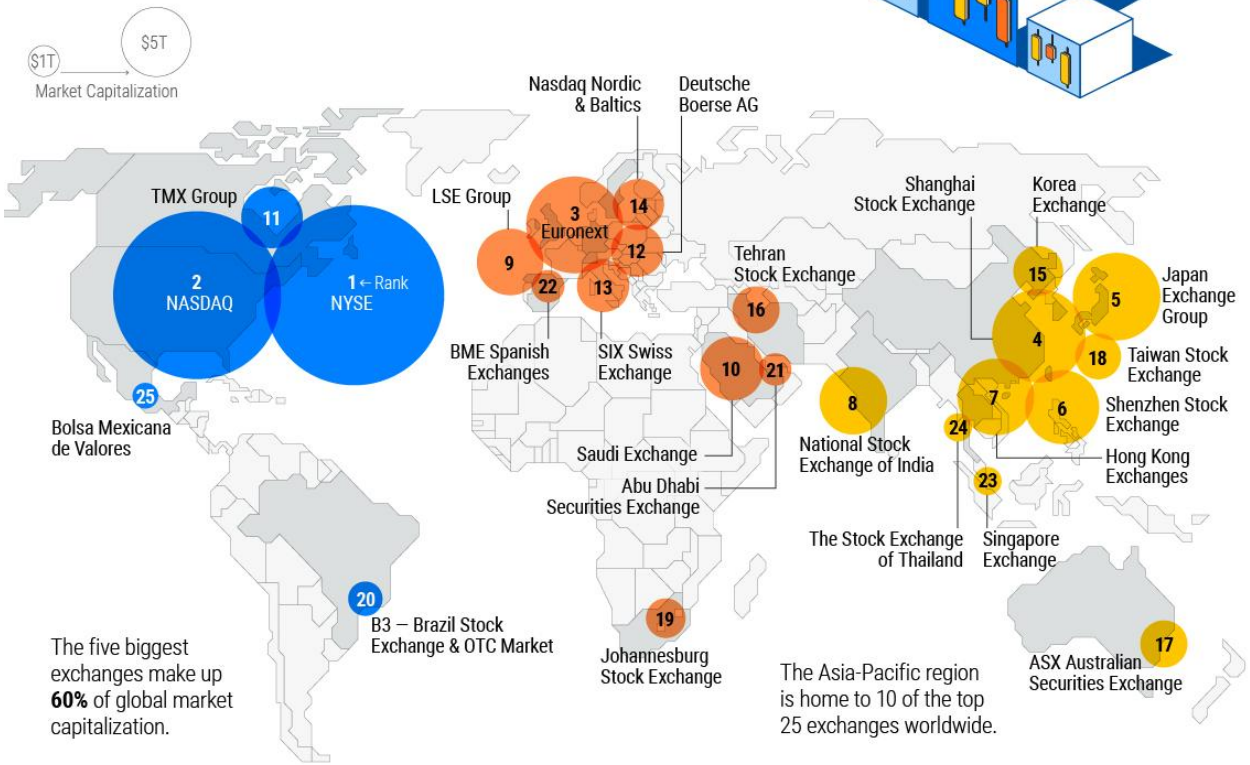
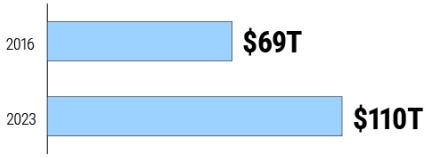
THE WORLD'S

Top 25 Stock Exchanges

Here are the largest stock exchanges in the world, spanning $106T in total market capitalization in 2023.



Total Market Value All Major Stock Exchanges



Data as of August 2023.
Source: World Federation of Exchanges

Market Share All Major Stock Exchanges

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Investing in Market Index

- Index funds:
  - Many mutual fund companies offer funds that invest in these portfolios (MSCI EMU, MSCI Europe, S&P 500, Wilshire 5000)
- Exchange-traded funds (ETFs):
  - Trade directly on an exchange but represent ownership in a portfolio of stocks.
    - Example: SPDRS (Standard and Poor's Depository Receipts) represent ownership in the S&P 500.

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The Market Risk Premium

- Estimation of the market risk premium:
  - $E[R_{Mkt}] - r_f$
- Determining the Risk-Free Rate
  - The yield on government bonds.
  - Surveys suggest most practitioners use 10 to 30 year government bonds.
- The Historical Risk Premium
  - Estimate the risk premium $(E[R_{Mkt}] - r_f)$ using the historical average excess return of the market over the risk-free interest rate.

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Historical Excess Returns

S&P 500 Excess Return Versus	Period 1926–2015	Period 1962–2015
One-year Treasury	7.7%	5.0%
Ten-year Treasury*	5.9%	3.9%

*Based on a comparison of compounded returns over a ten-year holding period.

Table 12.1 Historical Excess Returns of the S&P 500 Compared to One-Year and Ten-Year U.S. Treasury Securities

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The Market Risk Premium

- A fundamental approach
  - Using historical data has two drawbacks:
    - Standard errors of the estimates are large
    - Backward looking, so may not represent current expectations.
  - One alternative is to solve for the discount rate that is consistent with the current level of the index.

$$r_{Mkt} = \frac{Div_1}{P_0} + g = \text{Dividend Yield} + \text{Expected Dividend Growth Rate}$$

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Estimating Stock's Beta

- Estimating Beta from Historical Returns
  - Ideally, we would like to know a stock's beta in the future.
  - In practice we estimate it based on the stock's historical sensitivity.
  - Makes sense if a stock's beta remains relatively stable over time, which appears to be the case for most firms.
  - Typically data sources estimate correlations and volatilities from two to five years of weekly or monthly returns and use common market indices as the market portfolio.

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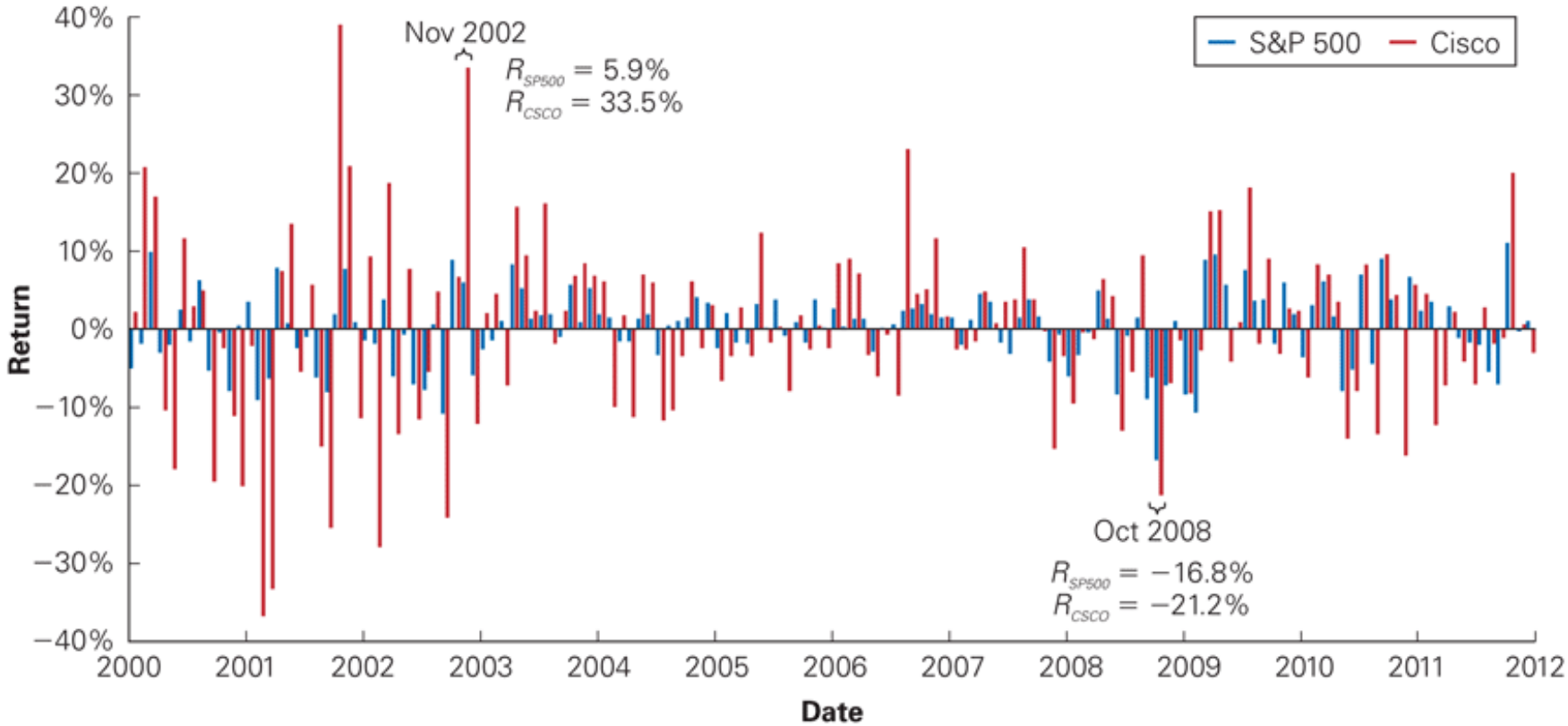
Estimating Stock's Beta



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Estimating Stock's Beta

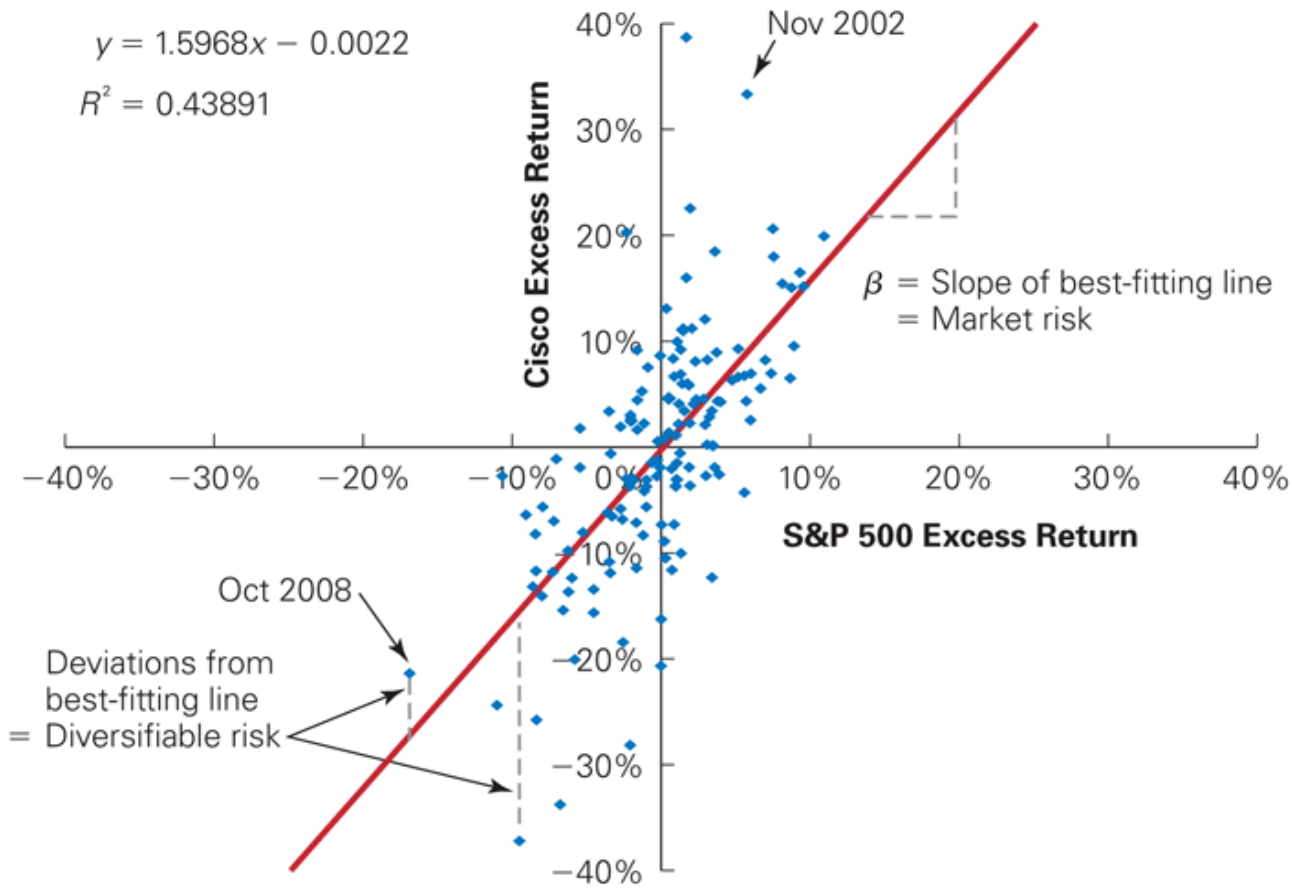

Figure 12.2 Scatterplot of Monthly Excess Returns for Cisco Versus the S&P 500, 2000-2012

Best Fitting Line

- As the scatterplot on the previous slide shows, Cisco tends to be up when the market is up, and vice versa.
- We can see that a 10% change in the market's return corresponds to about a 15% change in Cisco's return.
  - Thus, Cisco's return moves about two for one with the overall market, so Cisco's beta is about 1.5
- *Beta corresponds to the slope of the best-fitting line in the plot of the security's excess returns versus the market excess return.*
- Captures the components of a security's return that we can explain based on market risk.

Using Linear Regression

- Linear Regression
  - *The statistical technique that identifies the best-fitting line through a set of points.*
  - In terms of the excess return:

$$(R_i - r_f) = \alpha_i + \beta_i(R_{Mkt} - r_f) + \varepsilon_i$$

    - α_i is the intercept term of the regression.
    - $\beta_i(R_{Mkt} - r_f)$ represents the sensitivity of the stock to market risk. When the market's return increases by 1%, the security's return on average increases by β_i%.
    - ε_i is the error term and represents the deviation from the best-fitting line and is zero on average.

Using Linear Regression

- Since $E[\varepsilon_i] = 0$

$$E[R_i] = \underbrace{r_f + \beta_i(E[R_{Mkt}] - r_f)}_{\text{Expected return for } i \text{ from the SML}} + \underset{\text{Distance above / below the SML}}{\alpha_i}$$

- α_i represents a risk-adjusted performance measure for the historical returns.
  - If α_i is positive, the stock has performed better than predicted by the CAPM.
  - If α_i is negative, the stock's historical return is below the SML.

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Diversification with General Portfolios

- Given data for r_f, R_i, and R_{Mkt}, statistical packages for linear regression can estimate β_i.
  - A regression for Cisco using the monthly returns for 1996–2012 indicates the estimated beta is 1.597.
  - The estimate of Cisco's alpha from the regression is -0.22%.

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The Debt Cost of Capital

- Debt Yields
  - Yield to maturity: is the IRR an investor will earn from holding the bond to maturity and receiving its promised payments.
  - If there is little risk the firm will default, yield to maturity is a reasonable estimate of investors' expected rate of return.
  - If there is significant risk of default, yield to maturity will overstate investors' expected return.

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Debt's yield and expected return

- Consider a one-year bond with YTM of y:
  - For each $\$1$ invested in the bond today, the issuer promises to pay $\$(1+y)$ in one year.
- Suppose the bond will default with probability p, in which case bond holders receive only $\$(1+y-L)$
  - where L is the expected loss per $\$1$ of debt in the event of default.
- So the expected return of the bond is:

$$r_d = (1-p)y + p(y-L) = y - pL$$

$$= \text{Yield to Maturity} - \Pr(\text{default}) \times \text{Expected Loss Rate}$$

- The importance of the adjustment depends on the riskiness of the bond.

Annual Default Rates by Debt Rating (1983–2011)

Rating:	AAA	AA	A	BBB	BB	B	CCC	CC-C
Default Rate:								
Average	0.0%	0.1%	0.2%	0.5%	2.2%	5.5%	12.2%	14.1%
In Recessions	0.0%	1.0%	3.0%	3.0%	8.0%	16.0%	48.0%	79.0%

Source: "Corporate Defaults and Recovery Rates, 1920–2011," *Moody's Global Credit Policy*, February 2012.

- The average loss rate for unsecured debt is 60%.
- According to the Table, during average times the annual default rate for B-rated bonds is 5.5%.
- So the expected return to B-rated bondholders during average times is $0.055 \cdot 0.60 = 3.3\%$ below the bond's quoted yield.

Debt Betas

- Alternatively, we can estimate the debt cost of capital using the CAPM.
- But debt betas are difficult to estimate because corporate bonds are traded infrequently.
- Other means of estimating debt beta:
  - Chapter 21 shows a method for estimating debt betas using stock price data.
  - One approximation is to use estimates of betas of bond indices by rating category.

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Debt Betas

By Rating	*A and above*	*BBB*	*BB*	*B*	*CCC*
Avg. Beta	< 0.05	0.10	0.17	0.26	0.31
By Maturity	(BBB and above)	*1–5 Year*	*5–10 Year*	*10–15 Year*	> 15 *Year*
Avg. Beta		0.01	0.06	0.07	0.14

Source: S. Schaefer and I. Strebulaev, "Risk in Capital Structure Arbitrage," Stanford GSB working paper, 2009.

Average Debt Betas by Rating and Maturity

Example 1

- In early 2013, auto parts retailer Autozone had outstanding 10-year bonds with a yield to maturity of 3% and a BBB rating. If corresponding risk-free rates were 1.5%, and the market risk premium is 8%, estimate the expected return of Autozone's debt.
  - Remember that the average loss rate for unsecured debt is about 60%.
  - Calculate the cost of debt capital using two methods based on: default rate and CAPM

Rating:	AAA	AA	A	BBB	BB	B	CCC	CC-C
Default Rate:								
Average	0.0%	0.1%	0.2%	0.5%	2.2%	5.5%	12.2%	14.1%
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Example 1

- Default rates:
- $r_d = 3\% - 0.5\%(0.60) = 2.7\%$
- Alternatively, using CAPM
  - In that case,

$$r_d = 1.5\% + 0.10(8\%) = 2.3\%$$

- Both estimates are rough approximations and they both suggest that the expected return of Autozone's debt is below its yield-to-maturity of 3%.

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A Project's Cost of Capital

- A new project is not itself a publicly traded security!
  - We identify comparable firms in the same line of business.
  - Use the estimate of the cost of capital of the assets of comparable firms as a proxy for the project's cost of capital.
- All-equity as comparables:
  - Find an all-equity financed firm in a single line of business that is comparable to the project.
  - Use the comparable firm's equity beta and cost of capital as estimates

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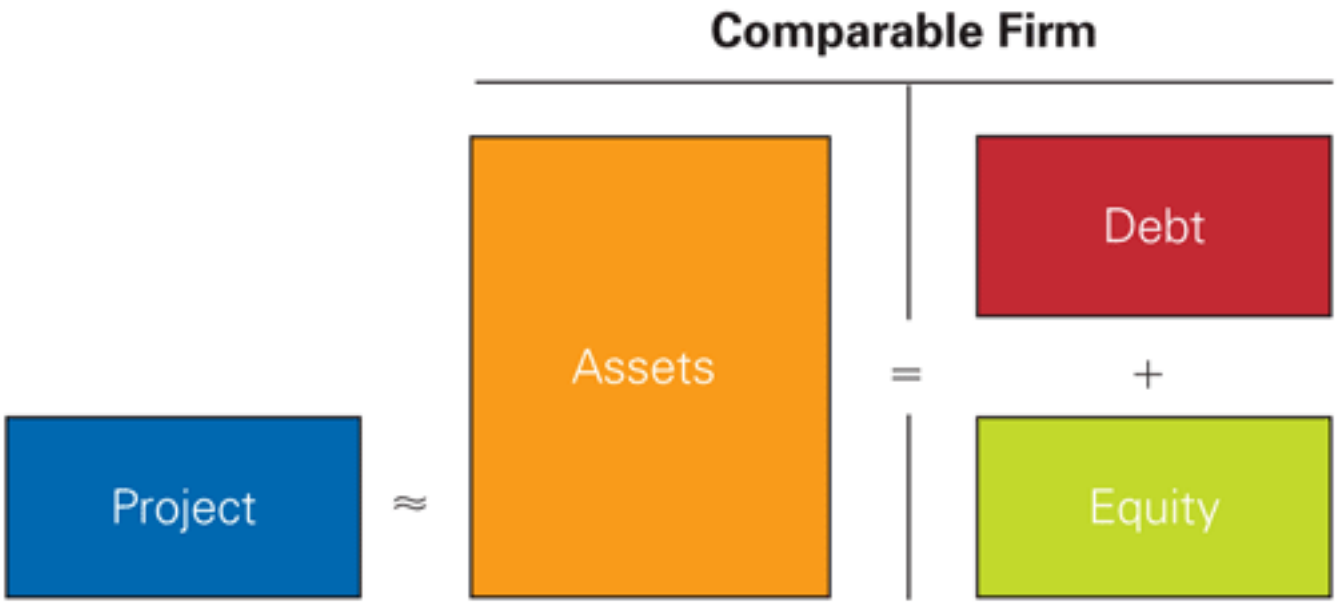
A Project's Cost of Capital

- Levered firms as comparables



Using a Levered Firm as a Comparable for a Project's Risk

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A Project's Cost of Capital

- Asset (unlevered) cost of capital
  - Expected return required by investors to hold the firm's underlying assets.
  - Equal to the weighted average of the firm's equity and debt costs of capital

$$r_U = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$$

  - Asset (unlevered) beta

$$\beta_U = \frac{E}{E+D} \beta_E + \frac{D}{E+D} \beta_D$$

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Cash and Net Debt

- Some firms maintain high cash balances.
- Cash is a risk-free asset that reduces the average risk of the firm's assets.
- Since the risk of the firm's enterprise value is what we're concerned with, leverage should be measured in terms of net debt.

Net Debt = Debt – Excess Cash and short-term investments

Industry Asset Betas

- We can combine estimates of asset betas for multiple firms in the same industry.
- Doing this will reduce the estimation error of the estimated beta for the project

Industry Asset Betas

Estimating an Industry Asset Beta

Problem

Consider the following data for U.S. department stores in mid-2009, showing the equity beta, ratio of net debt to enterprise value (D/V), and debt rating for each firm. Estimate the average and median asset beta for the industry.

Company	Ticker	Equity Beta	D/V	Debt Rating
Dillard's	DDS	2.38	0.59	B
J. C. Penney Company	JCP	1.60	0.17	BB
Kohl's	KSS	1.37	0.08	BBB
Macy's	M	2.16	0.62	BB
Nordstrom	JWN	1.94	0.35	BBB
Saks	SKS	1.85	0.50	CCC
Sears Holdings	SHLD	1.36	0.23	BB

Average Debt Betas by Rating and Maturity

By Rating	*A and above*	*BBB*	*BB*	*B*	*CCC*
Avg. Beta	< 0.05	0.10	0.17	0.26	0.31
By Maturity	(BBB and above)	*1–5 Year*	*5–10 Year*	*10–15 Year*	> 15 *Year*
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Source: S. Schaefer and I. Strebulaev, "Risk in Capital Structure Arbitrage," Stanford GSB working paper, 2009.

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Industry Asset Betas

Solution

Note that D/V provides the fraction of debt financing, and $(1 - D/V)$ the fraction of equity financing, for each firm. Using the data for debt betas from Table 12.3, we can apply Eq. 12.9 for each firm. For example, for Dillard's:

$$\beta_U = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D = (1 - 0.59)2.38 + (0.59)0.26 = 1.13$$

Doing this calculation for each firm, we obtain the following estimates:

Ticker	Equity Beta	D/V	Debt Rating	Debt Beta	Asset Beta
DDS	2.38	0.59	B	0.26	1.13
JCP	1.60	0.17	BB	0.17	1.36
KSS	1.37	0.08	BBB	0.10	1.27
M	2.16	0.62	BB	0.17	0.93
JWN	1.94	0.35	BBB	0.10	1.30
SKS	1.85	0.50	CCC	0.31	1.08
SHLD	1.36	0.23	BB	0.17	1.09
				Average	1.16
				Median	1.13

The large differences in the firms' equity betas are mainly due to differences in leverage. The firms' asset betas are much more similar, suggesting that the underlying businesses in this industry have similar market risk. By combining estimates from several closely related firms in this way, we can get a more accurate estimate of the beta for investments in this industry.

Financing and the Weighted Average Cost of Capital

- How might the project's cost of capital change if the firm uses leverage to finance the project?

- Answer: in week 40

Final Thoughts on the CAPM

- There are a large number of assumptions made in the estimation of cost of capital using the CAPM.
- How reliable are the results?
  - The types of approximation are no different from those made throughout the capital budgeting process. Errors in cost of capital estimation are not likely to make a large difference in NPV estimates.
  - CAPM is practical, easy to implement, and robust.
  - CAPM imposes a disciplined approach to cost of capital estimation that is difficult to manipulate.
  - CAPM requires managers to think about risk in the correct way.

Summary of main formulas

- SML equation $E(R_i) = r_f + \beta_i \times \left(E(R_m) - r_f\right)$
- Link between SML and linear regression

$$E[R_i] = \underbrace{r_f + \beta_i(E[R_{Mkt}] - r_f)}_{\text{Expected return for } i \text{ from the SML}} + \underset{\text{Distance above / below the SML}}{\alpha_i}$$

- Two ways to calculate debt cost of capital
  - $r_d = \text{Yield to Maturity} - \Pr(\text{default}) \times \text{Expected Loss Rate}$
  - $r_d = r_f + \beta_d \times \left(E(R_m) - r_f\right)$
- Asset (unlevered) cost of capital

$$r_U = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$$

- Asset (unlevered) beta

$$\beta_U = \frac{E}{E+D} \beta_E + \frac{D}{E+D} \beta_D$$

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Investor Behavior and Capital Market Efficiency

- Chapter 13
  - 13.1 Competition and Capital Markets
  - 13.2 Information and Rational Expectations
  - 13.3 The Behavior of Individual Investors
  - 13.4 Systematic Trading Biases
  - 13.5 The Efficiency of the Market Portfolio
  - 13.6 Style-Based Anomalies and the Market Efficiency Debate
  - 13.7 Multifactor Models of Risk
  - 13.8 Methods Used in Practice

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Learning objectives

- Explain how investors' attempts to "beat the market" should keep the market portfolio efficient.
- Describe the effect of homogeneous expectations on a security's alpha.
- Explain why holding the market portfolio does not depend on the quality of an investor's information or trading skills.
- Understand what the CAPM requires about investors' expectations.
- Explain trading behavioral biases bias and familiarity bias.
- Discuss why uninformed investors trade too much.
- Assess how uninformed investors' behavior deviates from the CAPM in systematic ways.
- Review why investors, on average, earn negative alphas when they invest in managed mutual funds.
- Assess the strategy of an investor "holding the market."
- Explain how the choice of the market proxy may lead to non-zero alphas.
- Discuss how systematic behavioral biases may affect the efficiency of the market portfolio.
- Describe the Arbitrage Pricing Theory.
- Discuss the Fama-French-Carhart model.

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Stock's Alpha

- The difference between a stock's expected return and its required return according to the security market line is called the stock's alpha.

$$\alpha_s = E[R_s] - r_s$$

- When the market portfolio is efficient, all stocks are on the security market line and have an alpha of zero

Deviations from the Security Market Line

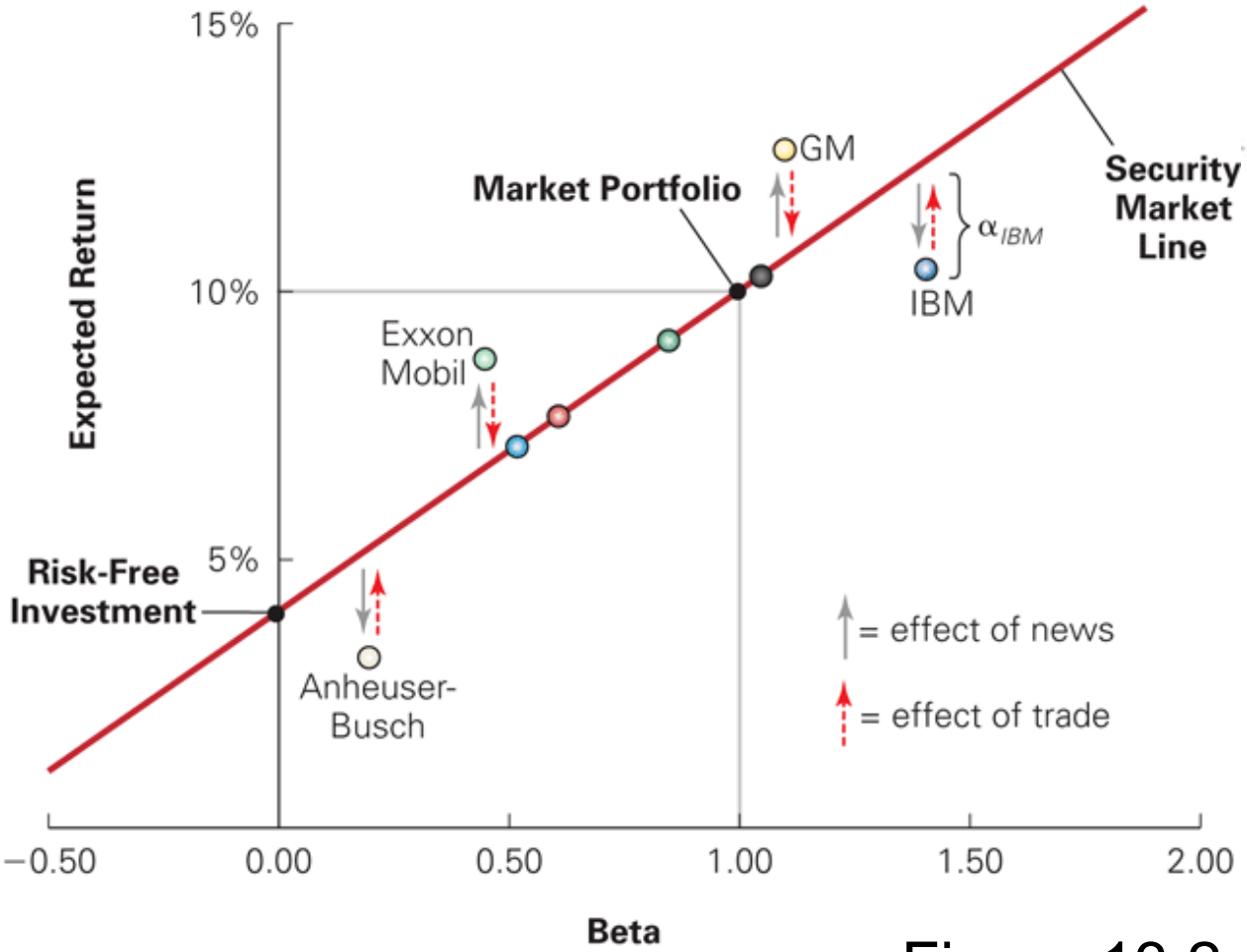


Figure 13.2

Profiting from Non-Zero Alpha Stocks

- Investors can improve the performance of their portfolios by buying stocks with positive alphas and by selling stocks with negative alphas.
- Two important consequences:
  - Savvy investors who try to beat the market should keep the market portfolio close to efficient much of the time.
  - There may exist trading strategies that take advantage of non-zero alpha stocks.

Information and Rational Expectations

- Informed Versus Uninformed Investors
  - In the CAPM framework, investors should hold the market portfolio combined with risk-free investments.
  - This investment strategy does not depend on the quality of an investor's information or trading skill.
  - Regardless of how much information an investor has access to, he can guarantee himself an alpha of zero by holding the market portfolio.

Rational Expectations

- Classic CAPM assumes homogeneous expectations (all investors agree on risk/return).
- In reality you can get the same result with weaker assumptions! What matters is that investors have **rational expectations**, they process their own information correctly and infer from prices.
  - Because the average portfolio of all investors is the market portfolio, the average alpha of all investors is zero.
  - If no investor earns a negative alpha, then no investor can earn a positive alpha, and the market portfolio must be efficient

Conclusion

- The market portfolio can be inefficient only if a significant number of investors either:
  - *Do not have rational expectations*: Misinterpret information and believe they are earning a positive alpha when they are actually earning a negative alpha,

  or

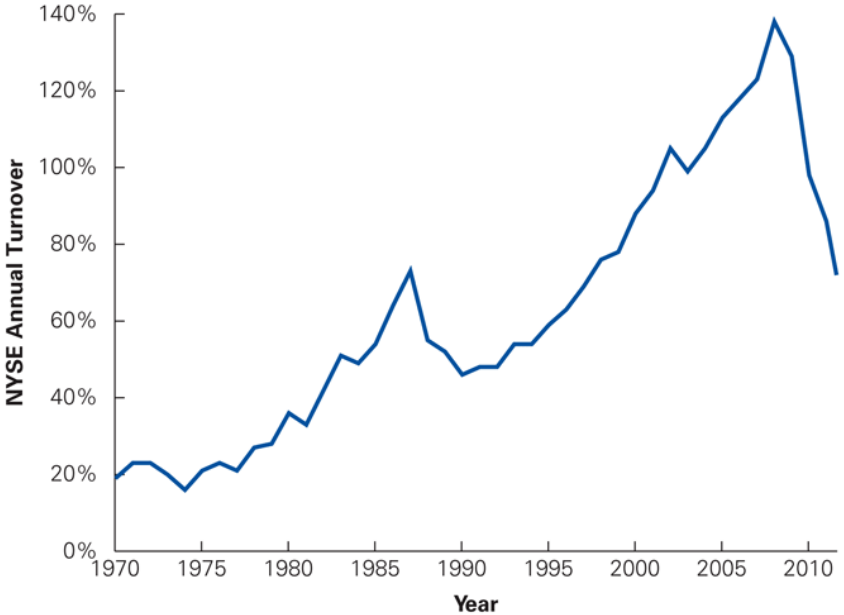
  - Care about aspects of their portfolios other than expected return and volatility, and so are willing to hold inefficient portfolios of securities.

The Behavior of Individual Investors

- Underdiversification and Portfolio Biases
  - There is much evidence that individual investors fail to diversify their portfolios adequately.
  - Familiarity Bias
    - Investors favor investments in companies they are familiar with
  - Relative Wealth Concerns
    - Investors care more about the performance of their portfolios relative to their peers.

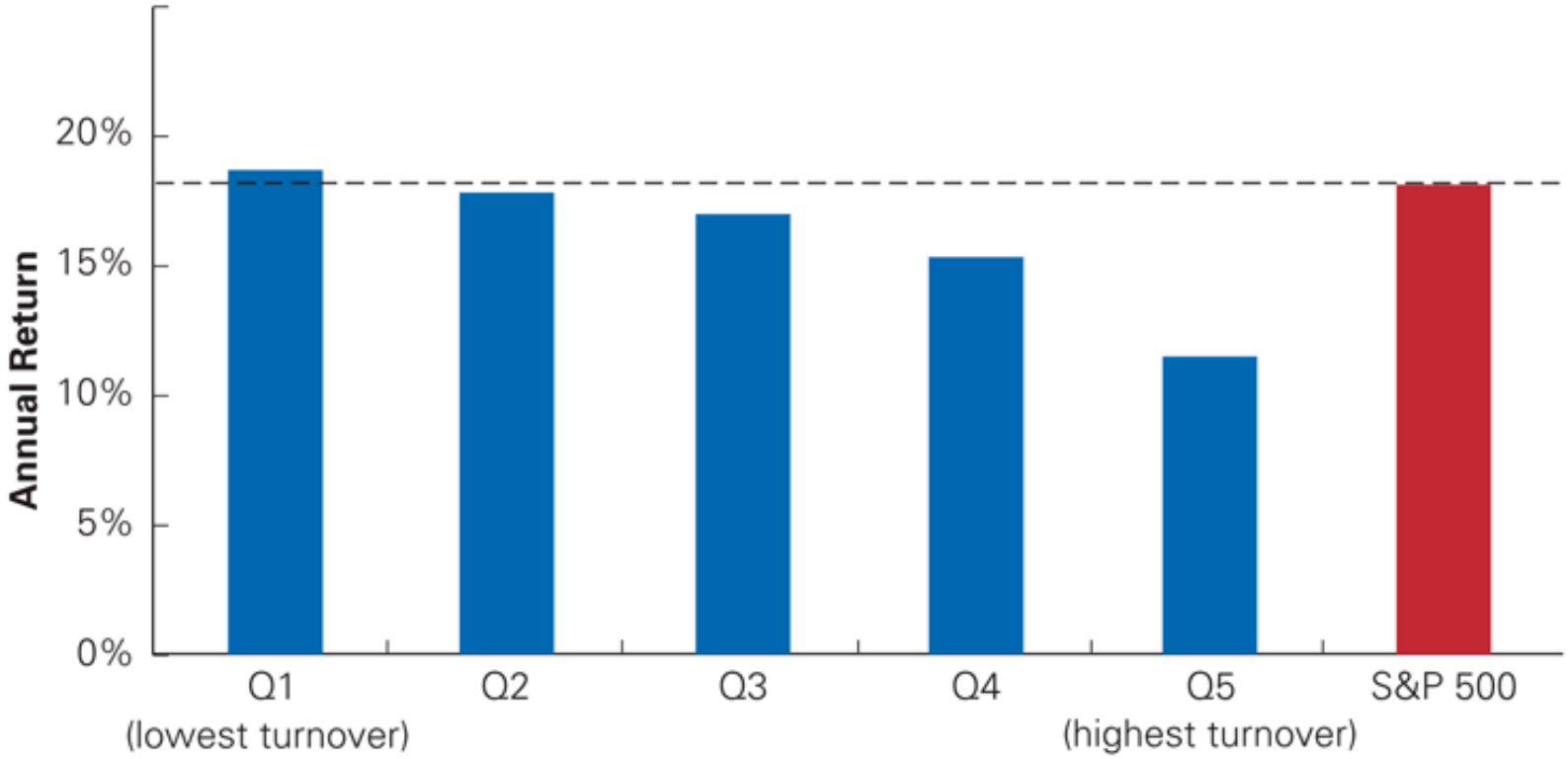
The Behavior of Individual Investors

- Excessive Trading and Overconfidence
  - According to the CAPM, investors should hold risk-free assets in combination with the market portfolio of all risky securities.
  - In reality, a tremendous amount of trading occurs each day.
- Possible Explanations:
  - Overconfidence Bias
  - Sensation Seeking



NYSE Annual Share Turnover, 1970–2011

Individual Investor Returns Versus Portfolio Turnover



Source: B. Barber and T. Odean, "Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors," Journal of Finance 55 (2000) 773–806.)

Individual Behavior and Market Prices

- If individuals depart from the CAPM in random ways, then these departures will tend to cancel out.
- Individuals will hold the market portfolio in aggregate, and there will be no effect on market prices or returns.
- In order to have an impact on the market, their must be patterns to the behavior of uninformed investors that are systematic.

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Systematic Trading Biases

- Hanging on to Losers and the Disposition Effect
  - When an investor holds on to stocks that have lost their value and sell stocks that have risen in value since the time of purchase
- Herd Behavior
  - When investors make similar trading errors because they are actively trying to follow each other's behavior
- Investor Attention, Mood, and Experience
  - Effects of news coverage, sunshine, past experiences

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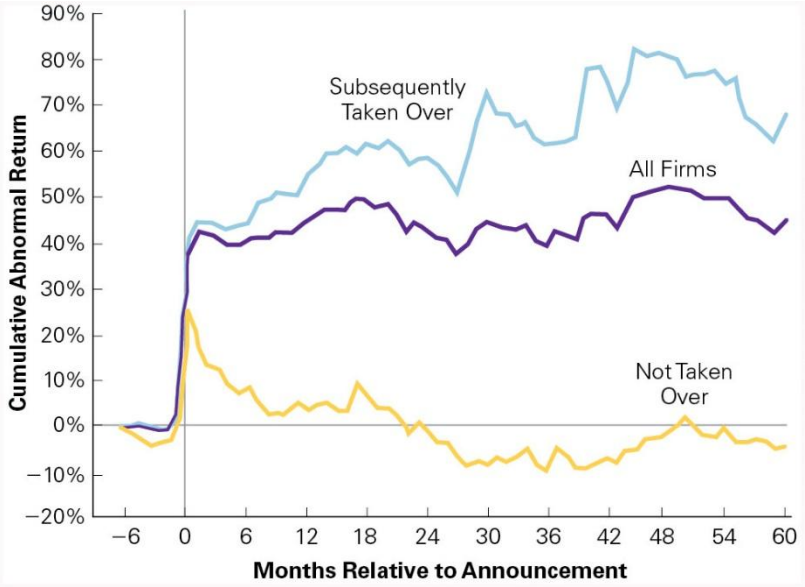
The Efficiency of the Market Portfolio

- Is there any evidence that investors can outperform the market without taking on additional risk?
- Trading on News or Recommendations
  - Takeover Offers: If you could predict whether the firm would ultimately be acquired or not, you could earn profits trading on that information.



The Performance of Fund Managers

- Fund Manager Value-Added
  - The median mutual fund actually destroys value
  - Most fund managers appear to trade so much that their trading costs exceed the profits from any trading opportunities they may find.
- Return to Investors
  - Numerous studies report that the actual returns to investors of the average mutual fund have a negative alpha
  - Superior past performance is not a good predictor of a fund's future ability to outperform the market

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Warren Buffett's bet against the hedge fund industry

- In 2007, Warren Buffett entered a million-dollar bet with fund manager Protégé Partners that the S&P 500 would beat a basket of hedge funds over the next decade.
- His S&P 500 index fund compounded a 7.1% annual gain over 10 years, beating an average increase of 2.2% by the basket of funds selected by Protégé Partners.
- The prize money will go to Girls Inc. of Omaha.
- Buffett took issue with hedge funds' high fees and their promise of outperforming the market.

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The Winners and Losers

- The average investor earns an alpha of zero, before including trading costs.
- Beating the market should require special skills or lower trading costs
  - Because individual investors are likely to be at a disadvantage on both counts, the CAPM wisdom that investors should “hold the market” is probably the best advice for most people

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Style-Based Anomalies and the Market Efficiency

- Excess Return and Market Capitalizations (size)
  - Small market capitalization stocks have historically earned higher average returns than the market portfolio, even after accounting for their higher betas.
- Excess Return and Book-to-Market Ratio
  - High book-to-market stocks have historically earned higher average returns than low book-to-market stocks.
- Momentum Strategy
  - Buying stocks that have had past high returns and (short) selling stocks that have had past low returns

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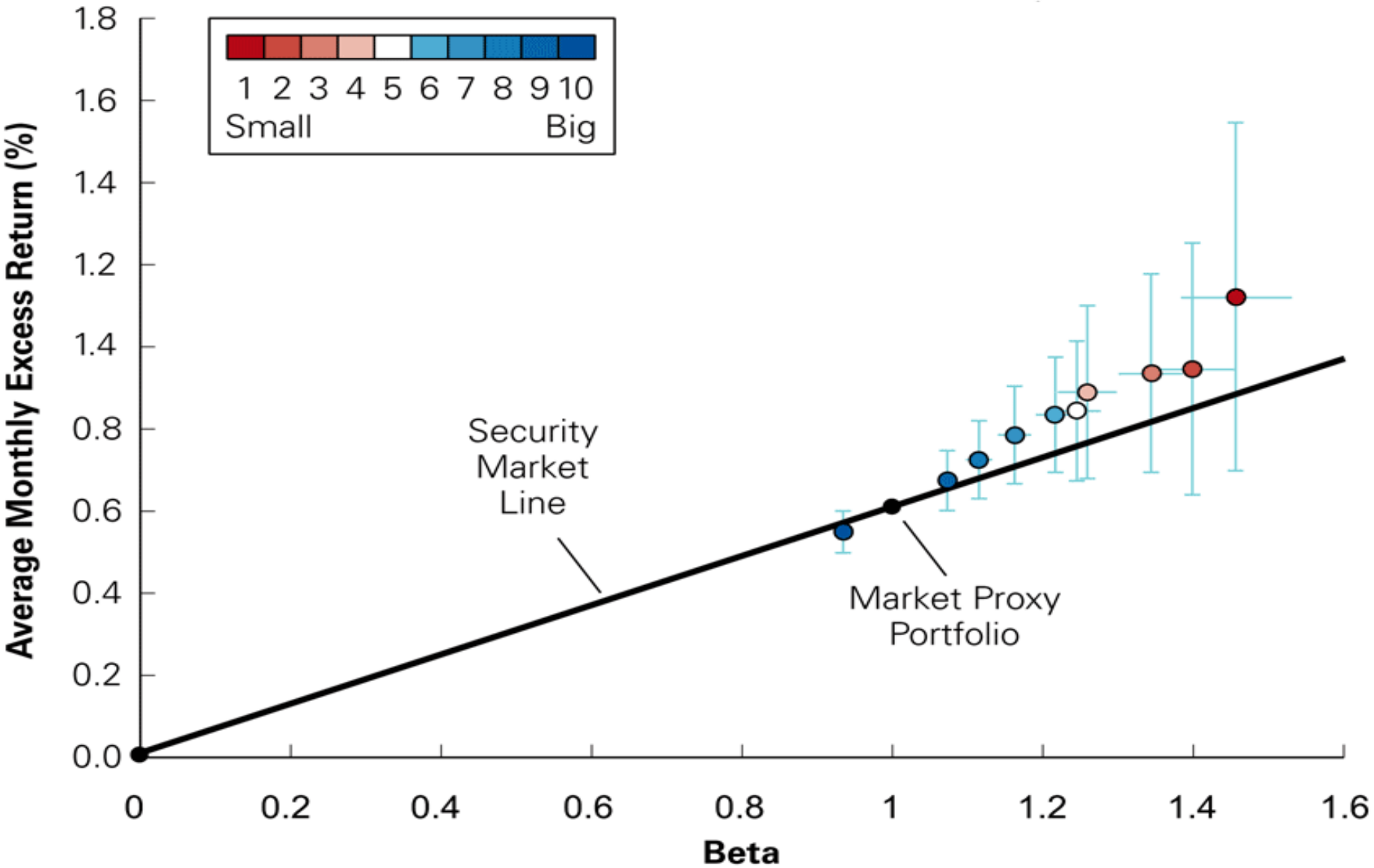
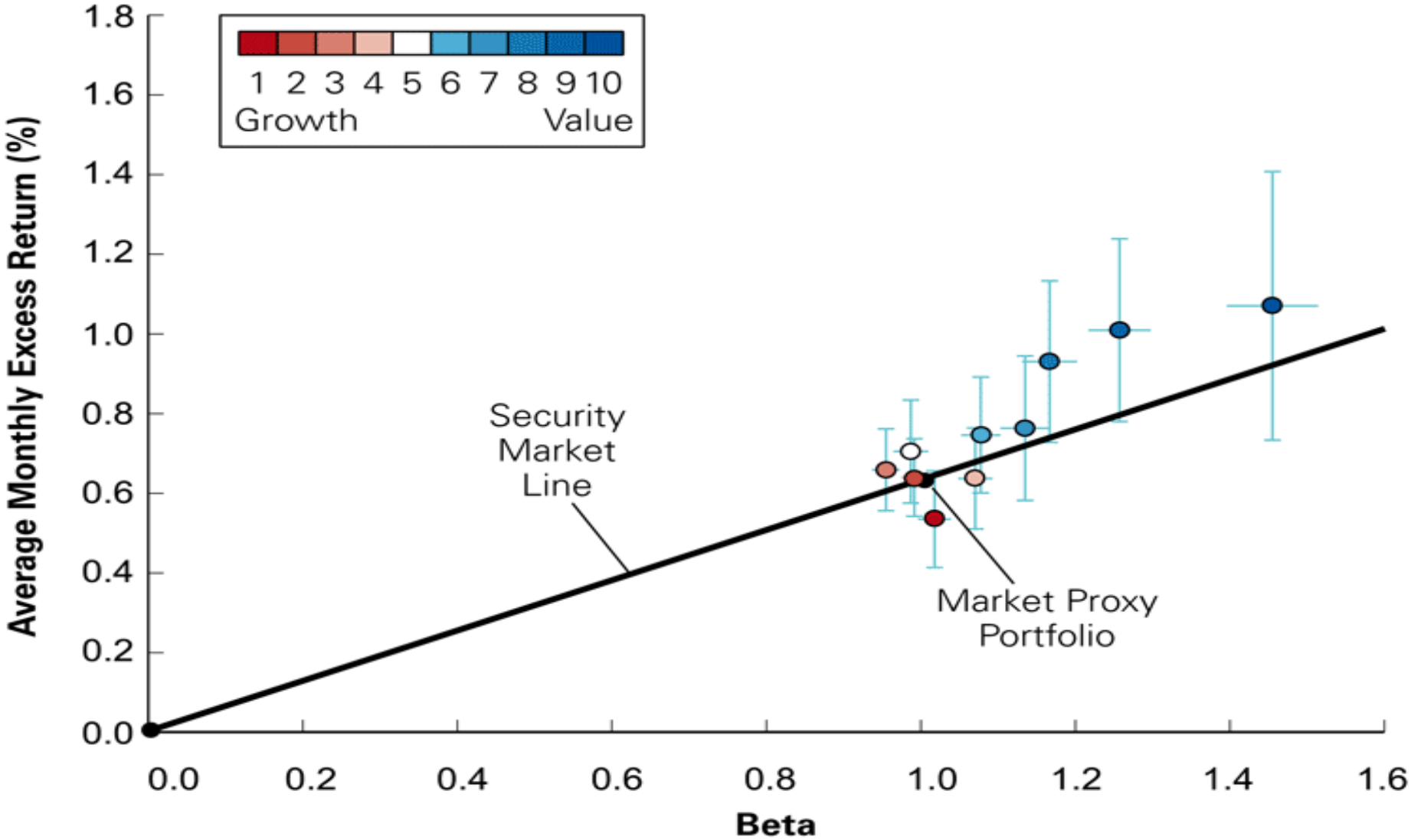
Excess Return of Size Portfolios, 1926–2008



Excess Return of Book-to-Market Portfolios, 1926–2008



Implications of Positive-Alpha Trading Strategies

- The only way positive-alpha strategies can persist in a market is if some barrier to entry restricts competition.
  - However, the existence of these trading strategies has been widely known for more than 15 years.
- Another possibility is that the market portfolio is not efficient, and therefore, a stock's beta with the market is not an adequate measure of its systematic risk.
  - Proxy Error
    - The true market portfolio may be efficient, but the proxy we have used for it may be inaccurate
  - Behavioral Biases
    - By falling prey to behavioral biases, investors may hold inefficient portfolios
  - Alternative Risk Preferences and Non-Tradable Wealth
    - Investors may choose inefficient portfolios because they care about risk characteristics other than the volatility of their traded portfolio.

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Multifactor Models of Risk

- The expected return of any marketable security is:

$$E[R_s] = r_f = \beta_s^{eff} \times (E[R_{eff}] - r_f)$$

- When the market portfolio is not efficient (do not capture all systematic risk), we have to find a method to identify an efficient portfolio before we can use the above equation.
- All that is required is to identify a collection of portfolios from which the efficient portfolio can be constructed.

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Using Factor Portfolios

- Given N factor portfolios with returns $R_{F1}, \ldots, R_{FN}$, the expected return of asset s is defined as

$$E[R_s] = r_f + \beta_s^{F1}(E[R_{F1}] - r_f) + \beta_s^{F2}(E[R_{F2}] - r_f) + \cdots + \beta_s^{FN}(E[R_{FN}] - r_f)$$

$$= r_f + \sum_{n=1}^{N} \beta_s^{FN}(E[R_{FN}] - r_f)$$

- $\beta^1 \ldots \beta^N$ are the factor betas

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Using Factor Portfolios

- Single-Factor Model
  - A model that uses one portfolio
  - The CAPM is an example of a single-factor model.
- Multi-Factor Model
  - A model that uses more than one portfolio in the model.
  - The Arbitrage Pricing Theory (APT) is an example of a multifactor model

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Using Factor Portfolios

- Self-financing portfolio:
  - can be constructed by going long in some stocks and going short in other stocks with equal market value
  - In general, a self-financing portfolio is any portfolio with portfolio weights that sum to zero rather than one

$$E[R_s] = r_f + \beta_s^{F1} E[R_{F1}] + \beta_s^{F2} E[R_{F2}] + \cdots + \beta_s^{FN} E[R_{FN}]$$

$$= r_f + \sum_{n=1}^{N} \beta_s^{FN} (E[R_{FN}])$$

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Selecting the Portfolios

- Self-financing portfolio that consists of a long position in the market portfolio that is financed by a short position in the risk-free security.
- Market Capitalization Strategy
  - A trading strategy that each year buys a portfolio of small stocks and finances this position by short selling a portfolio of big stocks has historically produced positive risk-adjusted returns.
    - This self-financing portfolio is widely known as the small-minus-big (SMB) portfolio.

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Selecting the Portfolios

- Book-to-market Ratio Strategy
  - A trading strategy that each year short sells an equally-weighted portfolio of stocks with a book-to-market ratio less than the 30th percentile of NYSE firms and finances this position by buying an equally-weighted portfolio of stocks with a book-to-market ratio greater than the 70th percentile of NYSE stocks has historically produced positive risk-adjusted returns.
  - This self-financing portfolio is widely known as the high-minus-low (HML) portfolio.

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Selecting the Portfolios

- Past Returns Strategy
  - Each year, after ranking stocks by their return over the last one year, a trading strategy that buys the top 30% of stocks and finances this position by short selling bottom 30% of stocks has historically produced positive risk-adjusted returns.
    - This self-financing portfolio is widely known as the prior one-year momentum (PR1YR) portfolio.
    - This trading strategy requires holding the portfolio for a year and the process is repeated annually.

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Multifactor Model Specification

- Collection of these four portfolios is currently the most popular choice for the multifactor model.
- This collection is referred to as the Fama-French-Carhart (FFC) factor specification.

$$E[R_s] = r_f + \beta_s^{Mkt}(E[R_{Mkt}] - r_f) + \beta_s^{SMB}E[R_{SMB}] + \beta_s^{HML}E[R_{HML}] + \beta_s^{PR1YR}E[R_{PR1YR}]$$

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Example

- You are considering making an investment in a project in the semiconductor industry.
- The project has the same level of non-diversifiable risk as investing in Intel stock.
- Assume you have calculated the following factor betas for Intel stock

$$\beta_{INTC}^{Mkt} = 0.171$$
$$\beta_{INTC}^{SMB} = 0.432$$
$$\beta_{INTC}^{HML} = 0.419$$
$$\beta_{INTC}^{PR1YR} = 0.121$$

- Determine the cost of capital by using the FFC factor specification if the monthly risk-free rate is 0.5%.

Solution

$$E[R_s] = r_f + \beta_s^{Mkt}(E[R_{Mkt}] - r_f) + \beta_s^{SMB} E[R_{SMB}]$$
$$+\beta_s^{HML} E[R_{HML}] + \beta_s^{PR1YR} E[R_{PR1YR}]$$
$$E[R_s] = 0.5\% + (0.171)(.61\%) + (0.432)(0.25\%)$$
$$+(0.419)(0.38\%) + (0.121)(0.70\%)$$
$$E[R_s] = .005 + .001043 + .001080 + .001592 + .000847$$
$$E[R_s] = .009562$$

- The annual cost of capital is .009562 × 12 = 11.47%

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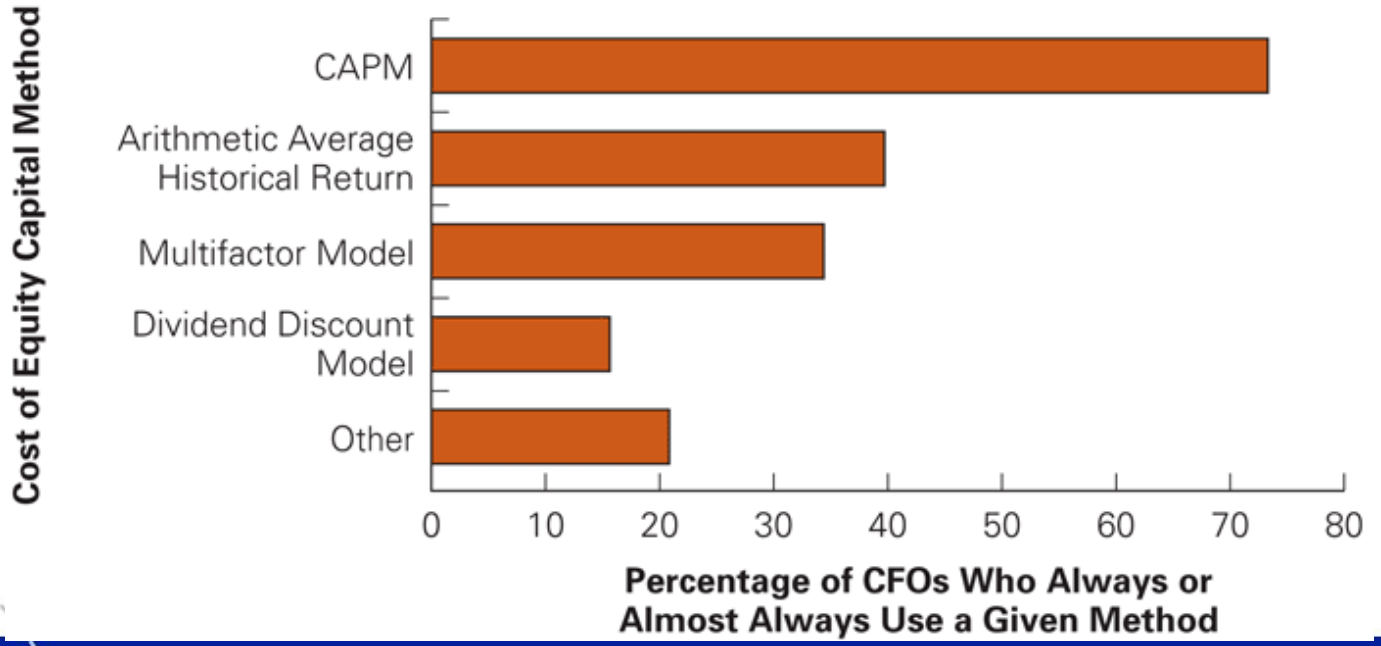
Fama-French-Carhart Factor Specification

- Although it is widely used in research to measure risk, there is much debate about whether the FFC factor specification is really a significant improvement over the CAPM
- One area where researchers have found that the FFC factor specification does appear to do better than the CAPM is measuring the risk of actively managed mutual funds

Methods Used In Practice

- There is no clear answer to the question of which technique is used to measure risk in practice. It very much depends on the organization and the sector.
  - There is little consensus in practice in which technique to use because all the techniques covered are imprecise.